Personnel Role Performance Audit — Slow Actions + Non-Realtime UI

Date: 2026-10-04 **Scope:** `src/app/personnel/` **and** `src/app/employee/` — action latency and list freshness, same criteria as the Super Admin audit **Companion:** `SUPER-ADMIN-PERFORMANCE-AUDIT.md` (§7 Solution applies 1:1 unless noted otherwise) **Repo:** `chikwaaa212/pgso`, branch `main`

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1. Executive Summary

The Personnel role suffers from **the same two diseases as Super Admin, plus one worse symptom and one bright spot**:

1. **Mutations block the UI on long server round-trips with no optimistic update.** Same `saving` spinner → `await action` → full `reload()` pattern on every write page.
2. **Lists go stale without refresh (same double-TTL stacking).** `useCachedAction` (30–60s, 10-min ceiling) over server Redis TTL (30–60s) on every list page.
3. **Worse: zero realtime anywhere.** Super Admin at least has 3 (broken) subscriptions; Personnel and Employee have **none** — verified by search. Two staff viewing the same queue diverge until TTL or manual refresh.
4. **Bright spot: personnel assets already has a paginated reader** (`assets/actions.ts:790-833`) with server-side filters and concurrent queries — the pattern the audit recommends — but Super Admin browse still uses the legacy heavy path.
5. **Mirror-image write-path tradeoff:** personnel busts cache fire-and-forget (faster response, stale-read race) while admin awaits the bust (slower response, correct read). Each side picked one half of the correct solution.

Issue	Super Admin	Personnel	Employee
Mutations feel slow (no optimistic UI)	Yes	Yes	N/A (read-only + request form)
Stale lists without refresh (double TTL)	Yes	Yes	Yes
Realtime subscriptions	3 wired, all broken (wrong table names)	0 wired	0 wired
Heavy list queries	Yes (`take:500` + per-row QR)	Yes, except assets has a paginated reader	Light
Cache bust blocks write response	Yes (awaited SCAN/DEL)	Mostly no (fire-and-forget → stale-read race)	N/A

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2. Finding P1 — No realtime on any personnel or employee page [CRITICAL]

Search for `useAdminRealtime` under `src/app/personnel/` **: no matches. Under** `src/app/employee/` : **no matches** (only `useCachedAction` in `employee/assets/page.tsx:11,51` and `employee/requests/page.tsx:11,37`).

Affected personnel pages (all TTL-only): dashboard (`dashboard/page.tsx:72`), deliveries (`deliveries/delivery-table.tsx:82`), inspections, requests (`requests/page.tsx:116`), assets, inventory (`inventory/page.tsx:112,128,133` — three hooks), issuances (`issuances/page.tsx:58`), repairs, documents (`documents/page.tsx:160,175,217` — three hooks), issues (`issues/page.tsx:51,65,71` — three hooks), logs (`logs/logs-table.tsx:56,70`).

Consequence: a delivery logged by Personnel A is invisible to Personnel B (and to Super Admin) until caches expire. There is no live convergence anywhere outside the 3 broken admin subscriptions.

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3. Finding P2 — Same double-TTL staleness [HIGH]

Identical layering as admin: client `useCachedAction` (`staleTime` 30–60s, `maxAge` 10 min, `sessionStorage` persistence) over server `withScopedCache` / `withCache` (30–60s Redis TTL). Fresh client cache skips the network entirely (`use-cached-action.ts:91-93`). Net effect: up to ~2 minutes of staleness by design on every personnel list, same as admin.

4. Finding P3 — Same pessimistic mutations [HIGH]

Personnel writes use the same blocking pattern. Representative: `personnel/requests/page.tsx:177-195` — after every file/approve/reject/complete/issue action, `reload()` busts **nine** client cache keys (dashboard, deliveries, inspections, inventory, assets, documents, issues, issuances, logs) and refetches. The user waits through the full server chain **plus** the refetch of heavy list queries before the UI settles. No page patches its rows optimistically first.

5. Finding P4 — Fire-and-forget bust: fast response, stale-read race [MEDIUM]

Personnel action modules wrap Next.js `revalidatePath` so every revalidation also busts Redis **without awaiting** (fire-and-forget):

```
// personnel/deliveries/actions.ts:8-13 (same helper in repairs, inventory, requests mod
function revalidatePath(path: string) {
  nextRevalidatePath(path)
  void import('@/lib/personnel-cache')
    .then((m) => m.bustPersonnelScopes(['deliveries', 'inspections', 'dashboard', 'docum
    .catch(() => {})
}
```

Effect: personnel actions respond faster than admin's (which `await`s the SCAN/DEL), but the client's immediate `refresh()` can land **before the bust completes** and re-cache the pre-write payload for another full TTL — the stale-read race from audit §4 B5. The codebase proves the team hit this: `personnel/requests/actions.ts:857-862` deliberately `await`s `bustPersonnelScopes` before reporting success, with a comment explaining why. The correct end state is neither: respond fast AND order correctly via `after()` (audit §7 Fix 3).

6. Finding P5 — Heavy list queries persist, but assets has the fix [MEDIUM]

Still full-table: `personnel/requests/actions.ts:436,448` (`take:500`), `personnel/repairs/actions.ts:347` (`take:500`), and the legacy `getAllUnifiedAssets` path (`asset take:500` + `inventoryItem take:500` , merged ~1000 rows with per-row QR generation — `assets/actions.ts:114-206,350-356`) which Super Admin browse still calls.

Bright spot: `assets/actions.ts:790-833` (`personnel:unified-assets-page` , 60s cache) already implements the recommended shape — server-side `page/q/status/source/condition/unit/category` filters, concurrent asset+stock queries via named promises + `Promise.all` , and lean `select` s (no QR in the list query). Remaining weakness: the `take:2000` cap is still a large payload; page sizes of 25–50 with a `{ rows, total }` return would finish the job (audit §7 Fix 4).

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7. Finding P6 — Cross-role gap, mirrored [MEDIUM]

Personnel writes bust only `personnel:*` scopes (`personnel-cache.ts` `bustPersonnelScopes`); admin writes bust only `super-admin:*` . Realtime — the mechanism that could paper over this gap — is dead on both sides. Lone exception: `personnel/requests/actions.ts:966` revalidates `/super-admin/requests` on one path, proving cross-role invalidation is possible but unsystematic.

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8. Solution Carry-Over (what to apply from §7, plus personnel-specifics)

The Super Admin §7 playbook applies unchanged; only the wiring list and one reuse item differ:

1. **Fix 0 + Fix 1 (realtime): add the hook to personnel pages** with the verified **plural** table names — `deliveries` (+ `inspections`), `inspections` (+ `deliveries`), `requests` , `assets` (+ `inventory`), `inventory` , `issuances` (`issuance_records`), `repairs` , `documents` , `issues` (`requests` , `repairs` — no dedicated model), `logs` (`audit_logs`), dashboard islands (`profiles`). Include the same 15s poll fallback and 700ms coalescing, and expose `refresh` as `onEvent` on every page (audit the personnel pages for the `deliveries` -style omission of `refresh`).

2. **Fix 2 (optimistic): replicate the overlay-map pattern** on personnel writes — log-delivery form (`log-delivery-mock-form.tsx:289` already busts client keys; add the pre-write row patch), request dialogs, inspection sign-off, issuance completion, inventory threshold cells.
3. **Fix 3 (write path): convert fire-and-forget busts AND awaited busts to `after()`** — one consistent shape: critical write awaited, audit + bust deferred. This resolves the personnel stale-read race and the admin latency in one move.
4. **Reuse, don't rebuild: point Super Admin browse at the paginated `unified-assets-page` reader** instead of legacy `getAllUnifiedAssets` , then tighten `take:2000` → real `{ rows, total }` pagination for both roles.
5. **Fix 5 (indexes): same DDL** benefits both roles since they share tables; add `Inspection.delivery_id` index after confirming the field name.

   Verify per role: two personnel browsers converging without refresh; mutation response after ~3 DB round-trips; publication check listing all 14 tables; no `take:500+` without pagination on either role's hot paths.

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9. Appendix — Evidence Index

- Zero realtime in personnel/employee: search `useAdminRealtime` under `src/app/personnel/` **and** `src/app/employee/` (no matches); employee `useCachedAction` at `employee/assets/page.tsx:11,51` , `employee/requests/page.tsx:11,37`
- Personnel cached lists: `personnel/dashboard/page.tsx:72` , `deliveries/delivery-table.tsx:82` , `inventory/page.tsx:112,128,133,149` , `requests/page.tsx:116,183` , `documents/page.tsx:160,175,217,270` , `issues/page.tsx:51,65,71` , `logs/logs-table.tsx:56,70` , `issuances/page.tsx:58` , `deliveries/log-delivery-mock-form.tsx:289`
- Pessimistic reload (9-key bust): `personnel/requests/page.tsx:177-195`
- Fire-and-forget bust helper: `personnel/deliveries/actions.ts:8-13` , `personnel/repairs/actions.ts:8-11` , `personnel/inventory/actions.ts:8-11` , `personnel/requests/actions.ts:8-11` ; awaited exception: `personnel/requests/actions.ts:857-862` ; cross-role exception: `personnel/requests/actions.ts:966`
- Heavy queries: `personnel/requests/actions.ts:436,448` , `personnel/repairs/actions.ts:347` , `personnel/assets/actions.ts:114-206,350-356`
- Paginated assets reader: `personnel/assets/actions.ts:790-833`
- Personnel write paths with extra scope busts: `personnel/deliveries/actions.ts:237-238,365-377` , `personnel/repairs/actions.ts:490-493,599-600,679-680` ,

`personnel/inventory/actions.ts:313,346,415-416`,
`personnel/issuances/actions.ts:927-933,1471-1477,1524-1547`,
`personnel/requests/actions.ts:1199-1200`

Audit of `main` following `SUPER-ADMIN-PERFORMANCE-AUDIT.md` Rev 2 criteria; no app code changed.